GENOMELENS

Economic analysis of a genomic report

Cost–utility model of acting on the findings in one whole-genome input, valued at $100,000 per QALY from a healthcare sector perspective.

Source input	**public benchmark genome huB4DBD7**	Input type	**WGS**
Generated	**24 October 2026**	Time horizon	**each decision model states its own**
Discounting	**3.0%, costs and QALYs**		

Proprietary.
Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.

Public benchmark genome.
A public benchmark genome. It is a real person's genome; the findings are shown to demonstrate the analysis and are not a clinical report.

Research and educational use — not medical or financial advice.
This is a decision-analytic model, not a clinical or economic evaluation submitted to any authority. Its parameters are population averages applied to one genome: it values the expected consequences of acting on a pattern of findings, and does not predict what will happen to any individual. Monetary figures include health valued at a willingness-to-pay threshold and are not cash returned to anyone. Clinically relevant findings require confirmation by an appropriate professional — a physician, clinical pharmacist or board-certified genetic counselor — before anything follows from them.

Economic impact overview

Modeled economic impact

NO HEADLINE VALUE IN THIS GENOME

No primary modeled value to headline: this genome's modeled values are shown below with their class, because they are measured nulls. Every other finding is classed with its reason in the table of all economically relevant findings.

1	4	2	2
TESTING-STRATEGY / POPULATION-LEVEL VALUE	TRUE ZERO	UNQUANTIFIED	TECHNICALLY UNRESOLVED

Modeled values (shown separately, never added)

TRUE ZERO EVIDENCE-QUALIFIED · TRUE ZERO

$0

CYP3A5 finding → CYP3A5-guided tacrolimus dosing. A measured null: the evidence finds no significant outcome difference

SUPPORTED CALCULATION AT THE ASSUMED AGE

Modeled net monetary benefit at assumed age 45: **$0** EVIDENCE-QUALIFIED · TRUE ZERO — the supported calculation at the assumed age: only decisions whose every model input is sourced (here a measured null, which does not depend on the age: the supported decisions show no measured effect on health, and no cost of acting is charged, which is not an unknown). A stricter test, not a total of the values above, and it does not replace them.

Research and educational use only; not medical, clinical or financial advice. Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results.

All economically relevant findings

Every economic result of this genome, ranked within its class. Values in different classes are never added.

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
Hypothetical, based on assumptions: any adult, before results	Population screening for Lynch syndrome, hereditary breast and ovarian cancer and familial hypercholesterolemia	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
CYP3A5 finding	CYP3A5-guided tacrolimus dosing	**$0**	TRUE ZERO EVIDENCE-QUALIFIED · TRUE ZERO	A measured null: the evidence finds no significant outcome difference
Factor V Leiden / Prothrombin carrier — situational anticoagulation		**valued at zero**	TRUE ZERO	A guideline recommends against intervening at this genotype, so care does not change
Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention		**valued at zero**	TRUE ZERO	The genotype does not change care
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)		**valued at zero**	TRUE ZERO	Not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified …
Alpha-1 Antitrypsin Deficiency carrier (codominant intermediate)		**not quantified**	UNQUANTIFIED	No disease model of its own yet
Nonsyndromic Cleft Lip / Palate carrier (uncertain)		**not quantified**	UNQUANTIFIED	No disease model of its own yet
CYP2C9 Intermediate Metabolizer (IM) (warfarin)		**not assessable**	TECHNICALLY UNRESOLVED	Warfarin dosing needs both CYP2C9 and VKORC1 called
NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing		**not assessable**	TECHNICALLY UNRESOLVED	NAT2 is inferred here from three markers (*5/*6/*7) without phase

All economically relevant findings (continued)

The evidence hierarchy

- **Evidence-supported**: every model input is sourced and the value does not rest on an age that is stated rather than observed (badge EVIDENCE-QUALIFIED)
- **Primary modeled**: the selected economic model for a decision; it may depend on the stated assumptions shown beside it
- **Once the decision arises**: the value per person if the decision arises; not an expected value today
- **Alternative modeled**: a competing model or sensitivity scenario for the same decision; never added to the primary result
- **Conditional**: valid only if a stated clinical or treatment condition holds; never added
- **Family / cascade-testing**: accrues to relatives or another beneficiary; never in the individual's personal total
- **Population / testing strategy**: a value per person screened or tested in a population, not this donor's value; never added **in this genome**
- **Hypothetical / projected**: an exploratory scenario on stated assumptions, not an observed or established benefit; in no total **in this genome**
- **Negative**: modeled, and the costs exceed the health gain valued at $100,000 per QALY; shown, never hidden
- **True zero**: the evidence supports no incremental benefit or change in care within the defined decision **in this genome**
- **Unquantified**: potentially relevant, but the evidence is insufficient for a monetary value; not $0 **in this genome**
- **Technically unresolved**: the genomic evidence cannot reliably support the interpretation the decision needs; not $0 **in this genome**
- **Not applicable**: the model or intervention does not apply to the assessed context (sex, age or setting); not $0

Unknown is not $0. Unquantified is not $0. Technically unresolved is not $0. No supported estimate is not a measured zero. Negative is not zero. An alternative is never added.

Findings valued by their decision models

Net monetary benefit (NMB) is the QALYs gained valued at $100,000 each, less the change in healthcare cost, in 2019 US dollars. It is a health-economic value, not cash saved: a positive NMB can come with higher spending. Each clinical decision is valued once; the categories are never added to each other. Each figure's horizon is its own model's (the carrier models run over a lifetime; no pooled disease model ran for this genome).

Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Age 45 is stated for this run, not observed (one demonstration age for every sample; the PGP public profiles give only an age bracket, and the GIAB and 1000 Genomes samples publish no age). A value whose model depends on age is modeled at that age; a value its source gives for one age only is held at that age, and its row says so. Model evidence is stated apart from whether its context matches the donor, and the qualification stays assumption-driven; an age-dependent value would be evidence-qualified only at an observed or self-reported age.

Sex: known male.

Supported calculation at assumed age 45

$0 EVIDENCE-QUALIFIED · TRUE ZERO · AGE 45 ASSUMED every decision supported at the assumed age is a measured null, so the supported calculation is a measured zero. Measured nulls eligible under the available donor context, each with its age restriction, are listed in A. Whether it applies to this donor is not established: the donor's age is not observed.

- no decision is supported at the assumed age beyond its measured nulls

A. Evidence-qualified expected value

$0 EVIDENCE-QUALIFIED · TRUE ZERO every evidence-qualified result is a measured null (a trial found no effect) or no change in care, so this is a measured zero, not an absence; the modeled values are in B.

- **CYP3A5-guided tacrolimus dosing**: $0 EVIDENCE-QUALIFIED · TRUE ZERO

  measured null in kidney transplant recipients (the trial: low immunological risk, living donors): the trial's pharmacokinetic primary (target concentration at day 3) was not improved (37.4% vs 35.6%, p = 0.79) and rejection, an underpowered secondary endpoint, did not differ (p = 0.82); a meta-analysis found no clinical-endpoint difference (acute rejection RR 1.00, 95% CI 0.64 to 1.55) although it did find more patients in the target range at steady state (RR 1.40, 95% CI 1.14 to 1.72), so no health gain and no cost of acting is credited. CPIC (strong) advises a 1.5 to 2 times starting dose for expressers because they are less likely to reach target concentrations, a drug-level rationale.

  comparator: standard dosing; sources: PMID 26714287, PMID 33751414.

B. Other modeled expected value, with further limitations

- none

C. Values that hold only in a stated clinical context

- none for this genome

D. Alternatives and scenarios, shown separately

- none for this genome

E. Actionable pathways not yet quantified

- Alpha-1 Antitrypsin Deficiency carrier (codominant intermediate) — evidence insufficient: no disease model of its own yet.
- Nonsyndromic Cleft Lip / Palate carrier (uncertain) — evidence insufficient: no disease model of its own yet.
- CYP2C9 Intermediate Metabolizer (IM) (warfarin) — technically unresolved: warfarin dosing needs both CYP2C9 and VKORC1 called; VKORC1 is not called in this input, so the warfarin decision is not priced.
- NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing — technically unresolved: NAT2 is inferred here from three markers (*5/*6/*7) without phase, which is not a prescribing-grade genotype, so no drug decision is valued. Missing input: a prescribing-grade NAT2 genotype.

Findings valued by their decision models (continued)

F. Valued at zero: care does not change (a result, not a gap)

- **Factor V Leiden / Prothrombin carrier — situational anticoagulation** — valued at zero: a guideline recommends against intervening at this genotype, so care does not change
- **Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention** — valued at zero: the genotype does not change care. No change in care (guideline): ACMG 2013, reaffirmed 2020: MTHFR polymorphism testing is not indicated; lowering homocysteine with B vitamins had no vascular benefit (PMID 23288205; PMID 32533132; PMID 20937919).
- **ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)** — valued at zero: not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified Benign in ClinVar (Variation 35707) and is a common polymorphism, not a Wilson disease allele (ClinVar Variation 35707).

Findings-first economics

No pooled result: no finding enters the pooled model, so there is no combined estimate to state; each finding is valued by its own decision model, by category, on the next page.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

HEALTH GAIN

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

HEALTHCARE COST CHANGE

Not available

no pooled result

MODELED NET BENEFIT

Not available

no pooled result

DECISION UNCERTAINTY

Not available

no pooled result

0

findings carry a standardized economic estimate

16 of the 16 findings shown carry no standardized estimate, and none is assigned one the model cannot support. Separately, 1 clinical decision(s) are valued by their own decision models (1 evidence-qualified, 0 modeled with stated assumptions), by category in the section 'Findings valued by their decision models'; a finding with such a value has its figure there, not here.

Folate Metabolism (MTHFR) · Alpha-1 Antitrypsin Deficiency… · Wilson Disease · Nonsyndromic Cleft Lip / Palate… · NAT2 — NAT2 slow acetylator… · CYP3A5 Intermediate expressor · CYP2C9 Intermediate Metabolizer · Factor V Leiden / Prothrombin… · CHRNA5 · Reproductive genetics · SERPINA1 carrier · MT-ND5 pathogenic · GJB2 pathogenic · PRSS1 pathogenic · BCHE pathogenic · SLC9B1 pathogenic

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype finding with a priced economic pathway; a finding whose decision the evidence values at zero is listed with the true zeros.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk and prevention findings, in detail

No risk or prevention finding of this genome has a standardized estimate, so there is no card to show. The findings are reported without one below, each with its reason on the glance pages; a lower evidence grade changes how a finding should be read, not whether it is shown.

Reported without a standardized value (15): Folate Metabolism (MTHFR)..., Alpha-1 Antitrypsin Deficiency carrier..., ATP7B rs1061472 (benign polymorphism)..., Nonsyndromic Cleft Lip / Palate carrier..., NAT2 — NAT2 slow acetylator..., CYP2C9 Intermediate Metabolizer (IM)..., and 9 more. Each carries its reason on the glance pages (no change in care, not assessable, unquantified, or reproductive information); no value is invented.

How the findings combine

Taken together, what do these findings mean?

There is no pooled reference case for this genome: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. Findings bearing on one condition are combined here before any value is credited; with nothing in the pool there is nothing to combine, and no figure stands in for the absent result. The findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'.

INCREMENTAL HEALTHCARE COST

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

INCREMENTAL QALYS

Not available

no pooled result

REFERENCE-CASE NMB

Not available

no pooled result

CLASSIFICATION

Not available

no pooled result

How sure is this, and what rests on judgment?

Uncertainty has two layers: the spread of modeled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across simulations

No probabilistic analysis of the pooled model: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. No probability of cost-effectiveness, interval or one-way swing is stated for it, because there is no pooled strategy to sample. The decision models carry their own parameter uncertainty (2,000 draws from the ranges the sources print, rescaled to a per-carrier value where a source prints another unit); where a decision has a range, its 5th to 95th percentile is printed beside that decision's value.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Base case: the value the published evidence supports for this genome, or “not estimable” with the reason. Scenario: what GenomeLens computes on its own assumptions, graded by the quality of its inputs, with each assumption numbered in the log below. Scenario figures are not from published data for this genome and are not the evidence headline; the reproductive costs belong to a prospective child and are in no personal total. A row valued by its own decision model repeats that model's value as a pointer; the model's basis (published, or hypothetical) is stated beside the headline. This genome's assessment is incomplete (a finding is actionable but unquantified, technically unresolved, or lacks the evidence to value): that is not a conclusion that the genome has no further economic value.

Table 1. Results by finding

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
This genome, to the person sequenced (each finding once)	**Not established** Incomplete assessment: not zero	**None projected** Sum of the rows not marked 'not summed': model and scenario values, never this donor's total	**no evidence-supported value; no value from decision models that are not evidence-qualified; no projected finding is summed (a placeholder on its own row is not added); figures marked 'not summed', reproductive values and payer values are excluded**
Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: the genotype does not change care Sources reviewed: PMID 17634462; PMID 32533132; PMID 35727271
Alpha-1 Antitrypsin Deficiency carrier (codominant intermediate)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: no value: the evidence does not support one: no disease model of its own yet
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: not a pathogenic variant
Nonsyndromic Cleft Lip / Palate carrier (uncertain)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: no value: the evidence does not support one: no disease model of its own yet
NAT2 — NAT2 slow acetylator — occupational cancer risk awareness	**Not estimable** Not estimable	**Not estimable** Conditional on a genotype call this input cannot establish; no figure is shown	No earlier figure; the decision models' verdict: not assessable from this input: NAT2 is inferred here from three markers (*5/*6/*7) without phase, which is not a … Sources reviewed, none of which carries a figure: PMID:16112301
CYP3A5 Intermediate expressor (IM) (tacrolimus)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Valued once, by the decision model(s) of this report: 'CYP3A5-guided tacrolimus dosing' (a measured null)
CYP2C9 Intermediate Metabolizer (IM) (warfarin)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: not assessable from this input: warfarin dosing needs both CYP2C9 and VKORC1 called; VKORC1 is not called in this input …

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
Factor V Leiden / Prothrombin carrier — situational anticoagulation	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: a guideline recommends against intervening at this genotype, so care does not change
CHRNA5 — Nicotine-dependence susceptibility variant	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care
Reproductive genetics — 3 carrier condition(s) where partner testing would be informative	**Separate arm** Separate arm	**See reproductive table** Reproductive table	Costed per risk in the reproductive rows (a prospective child's value, kept out of the personal total)
SERPINA1 carrier — heterozygote health	**Not estimable** Not estimable	**No model yet** No model yet	Heterozygote health effects are real but not costed per gene
MT-ND5 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
GJB2 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
PRSS1 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
BCHE pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
SLC9B1 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
F5 Factor V Leiden (R506Q) Venous Thromboembolism (VTE) Susceptibility; dominant; carrier	**1 in 2** that a child inherits it, per pregnancy	**$10,000** curated outcome cost for a child with one copy	**Not stated** per pregnancy	**$185** a known-variant test of the pregnancy or the child	GenomeLens registered estimate
SERPINA1 Z allele (Glu342Lys) Alpha-1 Antitrypsin Deficiency; recessive; carrier	**1 in 200** of an affected child, per pregnancy	**$531,062** lifetime healthcare cost of the condition, present value	**$2,655** per pregnancy	**$387** the partner's carrier test (expanded panel)	PMID:30775420

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European) (continued)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
ATP7B K832R Wilson Disease (variable); recessive; common polymorphism, not a disease allele	**not stated** of an affected child, per pregnancy	**Not available** this allele is commonly reported as a polymorphism rather than a disease allele (classified as uncertain significance in the carrier catalog): it cannot complete an affected genotype, so neither a chance of an affected child nor a cost of the child's condition is stated; there is no risk for a partner's test to resolve, so none is priced	— per pregnancy	— No test priced: no risk to resolve	GenomeLens registered estimate
IRF6 Cleft lip / palate Nonsyndromic Cleft Lip / Palate; undeclared; carrier	**1 in 2** that a child inherits it, per pregnancy	**Not available** no cost of the child's condition is registered for IRF6; only the test that resolves the risk is priced	— per pregnancy	**$185** a known-variant test of the pregnancy or the child	GenomeLens registered estimate

Methods

How these figures were produced, and how to read them

A decision-analytic model values the expected consequences of acting on each finding in this genome. The figures are model outputs under stated assumptions, kept apart from clinical outcomes.

ANALYTIC FRAME

Perspective	**healthcare sector**
Comparator	**each decision model states its own**
Outcomes	**cost, QALYs, NMB**
Time horizon	**each decision model states its own**
Discounting	**3%, costs and QALYs**
Willingness to pay	**$100,000/QALY**
Currency	**USD, United States**
Uncertainty	**decision models: 2,000-draw parameter uncertainty; pooled model: no PSA (no pooled model)**

HOW TO READ EACH FIGURE

- **Evidence-supported value:** every model input is sourced and the value does not rest on an age that is stated rather than observed. Where the age is stated, as for every public genome, an age-dependent value is a standardized scenario and is labeled so.
- **Decision-model rows:** computed from published inputs with at least one assumption (for a public genome, the stated age); shown, never presented as evidence-qualified.
- **Hypothetical, based on assumptions:** what a finding could be worth if its curated assumptions held; never added to any counted total.
- **Projected, based on assumptions:** the GenomeLens scenario beside a base case that cannot be computed soundly; not the evidence headline.
- **Reproductive costs:** per pregnancy, for a prospective child, never in a personal total.

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labeled as such.
- Costs are restated to 2019 US dollars from each source's price year.
- Findings without a modeled decision pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Proprietary method. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.